Karim Abdel Sadek

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EDUCATION

University of Amsterdam **Amsterdam, NL**
MSc in Artificial Intelligence *Sep 2023 - June 2025*

- Relevant Coursework: Machine Learning I, Deep Learning II, Game Theory, Knowledge Representation
- GPA: 8.5 / 10 (Equivalent to 4.0/4.0)

Georgia Institute of Technology **Atlanta, USA**
BSc in Computer Science – Exchange student *Jan 2023 – May 2023*

- Selected by academic merit to participate in the Exchange Program 2022-2023 with a full ride scholarship.
- Coursework: Deep Learning, Information Theory, Quantum Mechanics, Automata and Complexity
- GPA: 4.0/4.0

Bocconi University **Milan, IT**
BSc in Mathematical and Computing Sciences for Artificial Intelligence *Sep 2020 – July 2023*

- Relevant Coursework: Advanced Programming, Linear Algebra, Machine Learning, Advanced Mathematical Analysis, Algorithms and Data Structures, Game Theory, Optimization.
- Thesis: '*Machine `Learning Augmented Algorithms with Reduced Number of Predictions*', grade: 4/4
- Head Student, Course Representative, Student-Faculty Joint Teaching Committees Representative
- GPA: 3.7/4.0

WORK EXPERIENCE

University of Cambridge – Kruger AI Safety Lab **Remote**
Research Intern *Feb 2024 –Sep 2024 (Exp.)*

- Working on Goal Misgeneralization and Specification Misgeneralization in RL
- Supervisors: Usman Anwar, Michael Dennis, David Krueger

Bocconi University, Department of Computing Sciences **Milan, IT / Remote**
Theoretical Computer Science Student Researcher *Jun 2022 – Jan 2024*

- First co-author on a paper titled 'Algorithms for Caching and MTS with a reduced number of predictions'
- Research Area: Theory of Algorithms, ML-augmented algorithms, Online learning
- We propose new ML-augmented algorithms for Metrical Task Systems and caching, with an emphasis on optimizing performance through ML predictions while ensuring consistency, robustness, and smoothness, using the predictions parsimoniously.
- Paper has been accepted at ICLR 2024
- Supervisor: Prof. Marek Elias

Clearbox.AI **Turin, IT**
Machine Learning Intern *Jun 2022 – Aug 2022*

- Worked on the generation of synthetic data of single-table and multi-table datasets for internal and business purpose
- Benchmark of the SDV MIT library against the internal privacy metrics of the company, analyzed the results through advanced statistical methods
- Synthesized tens of dataset using both GANs and VAEs. Increased the metrics score on synthetic generated dataset by 21%

Bocconi University **Milan, IT**
Teaching Assistant *Feb 2022 – Jan 2023*

- I have been a TA for the course 'Algorithms and Computing Theory' taught by Luca Trevisan.
- Held weekly office hours and designed exercise sessions to improve student performance

PUBLICATIONS

Abdel Sadek Karim., Elias Marek., '*Algorithms for Caching and MTS with reduced number of predictions*', *International Conference on Learning Representations (ICLR). 2024.* URL: Paper and Reviews

Abdel Sadek Karim., Nulli, M., Velja, J., Vincenti J., '*Explaining RL Decisions with Trajectories: A Reproducibility Study*', *Paper Under Review.*

HONORS, AWARDS and SCHOLARSHIPS

University of Oxford, Bioinformatics Hub Conference Series **Oxford, UK**
ML Group Competition Winner, Student Presenter *Jul 2022*

- Winner of the AI LAB competition held by Prof. Francesca Buffa regarding breast cancer detection
- Designed an end-to-end ML pipeline, working on 4 datasets each consisting of 30k+ datapoints and 200+ features. Our team built 4 classifiers reaching a 96% accuracy in detecting hypoxia
- Received funding to present our work at one conference of the Bioinformatics Hub Conference Series organized by the Department of Oncology of University of Oxford.

Lead The Future Mentorship **Italy**
Mentee *Sep 2022 - Present*

- Among the few Italian students selected to be mentees for Lead The Future, a leading mentorship non-profit organization for students in Stem, with an acceptance rate below 20%

Scholarships and Grants

- Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements , €300 2014
- AFS Intercultura: AFS Scolarship for Academic Excellence. Covered the cost of my exchange year in Brazil for the academic 2018/2019, ~€4000 2018
- Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements , €500 2020
- Bocconi University: Funding to present the AI LAB winning project at Oxford University, ~€600 2021
- Georgia Institute of Technology: Full-ride scholarship, Spring ’23, ~$15,000 2023
- Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements , €600 2023
- University of Amsterdam: MSc Student Travel Grant for ICLR 2024, €500 2024
- Qualcomm Technologies and Ellis Amsterdam: MSc Grant, €2000 2024
- University of Toronto and CIFAR: AI Inclusive Scholarship for DLRL, ~ $2900 2024
- Krueger AI Safety Lab and ERA Fellowship: Research Intern Summer ’24, ~ $14,000. 2024

LANGUAGES & SKILLS

Languages: Italian: Native | English: Advanced, IELTS 8.0 | Portuguese: Advanced, C1 | Spanish: Good, Dele B1

Skills: Advanced proficiency in Python |Tensorflow, PyTorch, Jax | R | C

EXTRACURRICULARS

Student Associations:

- Head of Research at Bocconi Students for Data Science and AI
- Member at Bocconi Students for Space Developments

Volunteering:

- Teacher Volunteer: Taught math and physics to final year high-school students in a difficult context in Brazil
- AFS Volunteers: I helped and mentored people who will spend a school period abroad to develop the social and cultural skills, and sensibility needed for such life experience. I gave them advice speaking about my experience in Brazil. Prepared over 60 students.